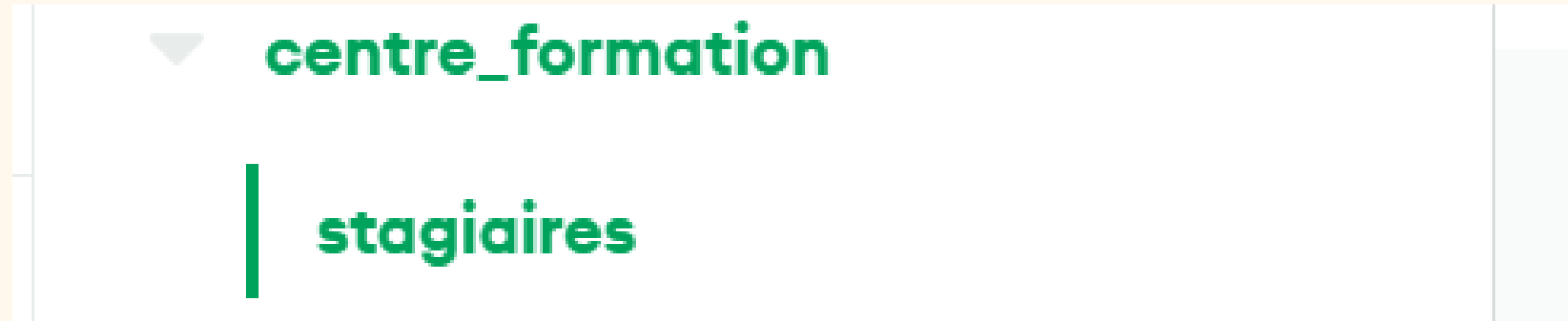


RAPPORT

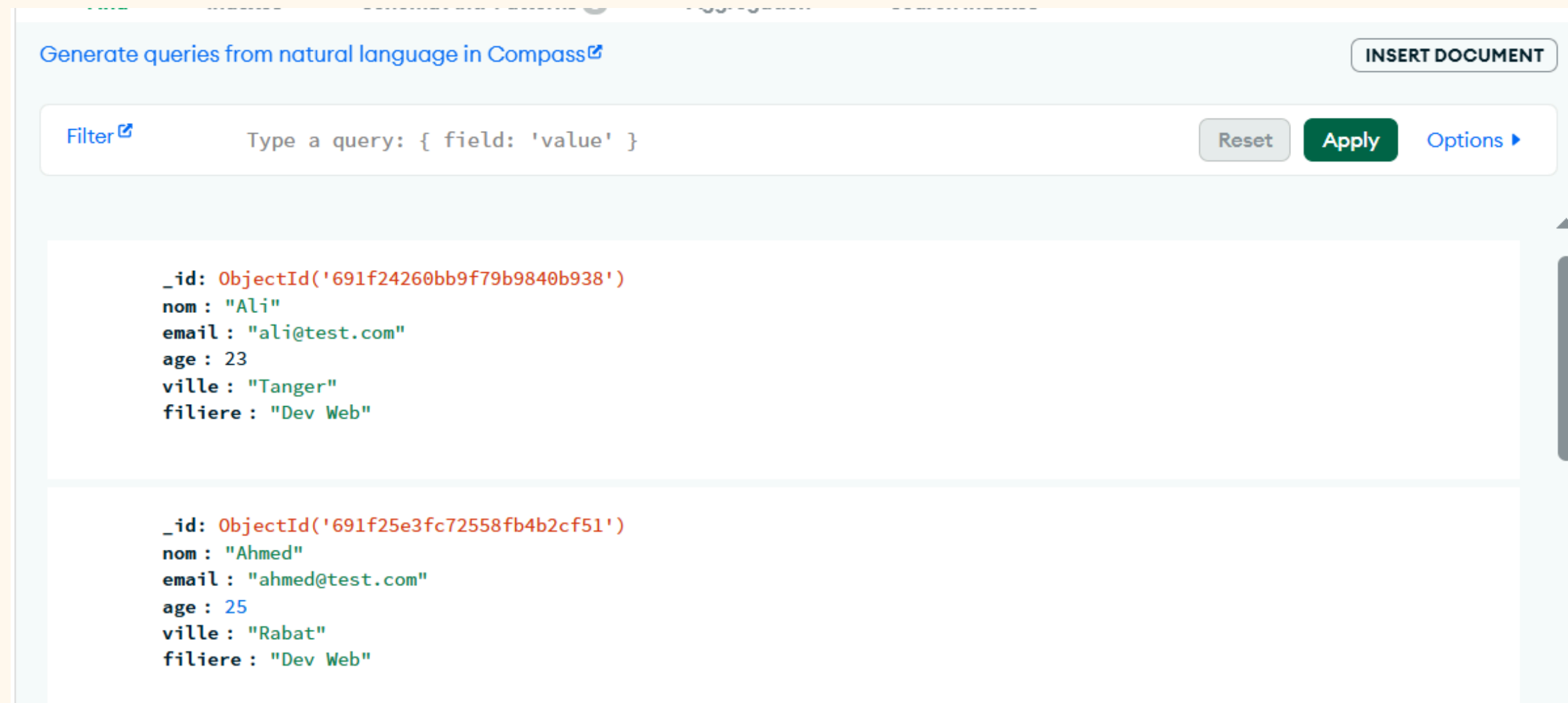
ATELIER 7 – APPLICATION GRAPHIQUE
PYTHON + MONGODB ATLAS “BACK-
OFFICE DES STAGIAIRES DU CENTRE
DE FORMATION”

Partie 1 : “Back-office des stagiaires du centre de formation”

Une base MongoDB Atlas : **centre_formation** et Une collection : **stagiaires**



Insertion des documents ;



Configuration MongoDB Atlas:

+ ADD NEW DATABASE USER

User ↕	Description	Authentication Method ↕	MongoDB Roles	Resources	Actions
 chaimaechabouti_db_user		SCRAM	atlasAdmin@admin	All Resources	EDIT DELETE
 python_user		SCRAM	readWriteAnyDatabase@admin	All Resources	EDIT DELETE

Partie 2 : Tableau de Bord Statistique

config.py

```
1 MONGODB_URI = "mongodb+srv://python_user:MotDePasseFort12345@cluster0.za2xhpm.mongodb.net/"
2 DB_NAME = "centre_formation"
```

requirements.txt

```
1  pymongo
2  dnspython
3  matplotlib
```

gui_atlas.py

Gestion des Stagiaires - Centre de Formation

Nom*

Email*

Âge

Ville

Filière

+ Ajouter Stagiaire

✎ Modifier

🗑 Supprimer

📊 Dashboard

Filtrer par ville:

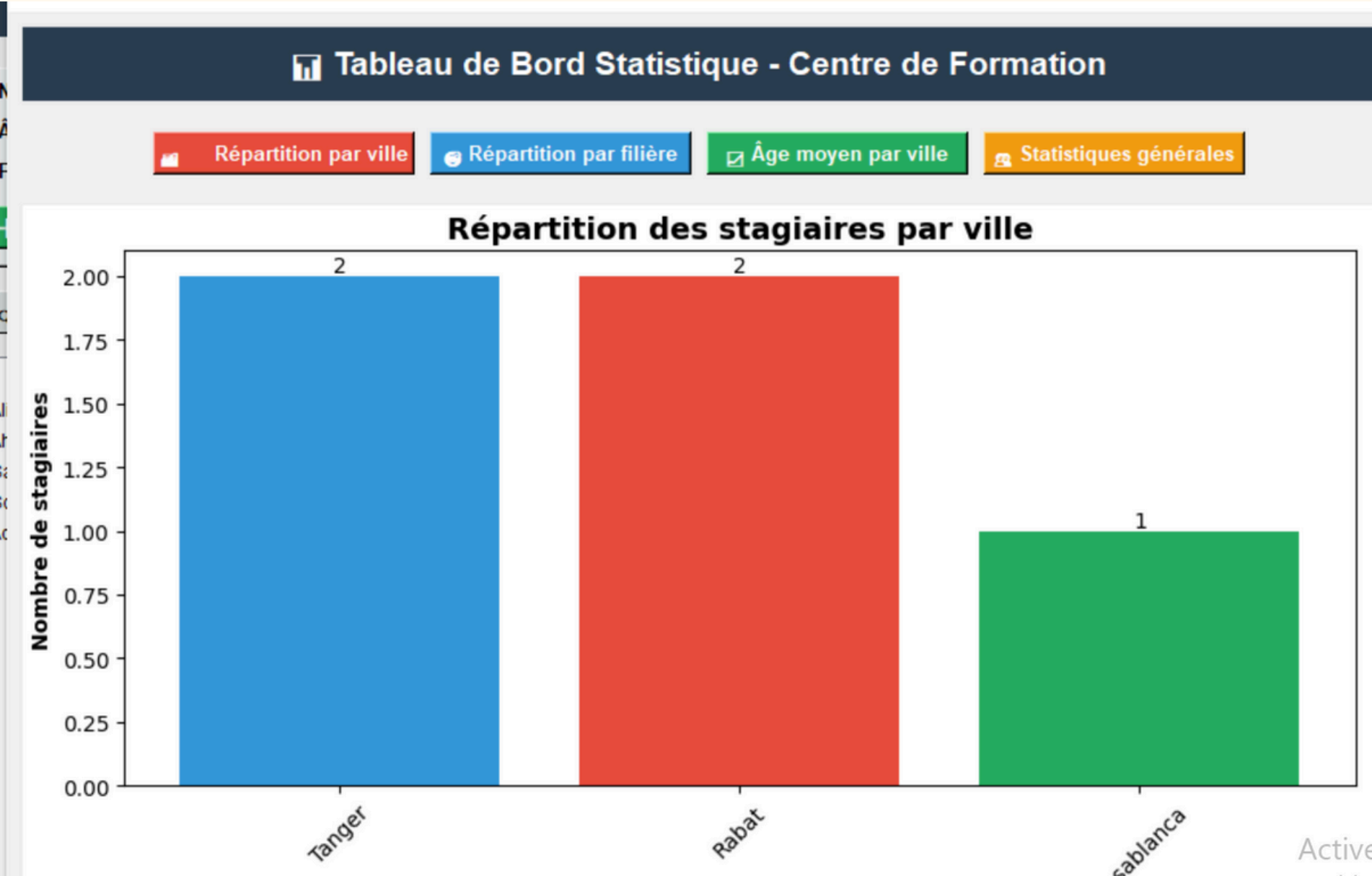
(Toutes)

Rafraîchir

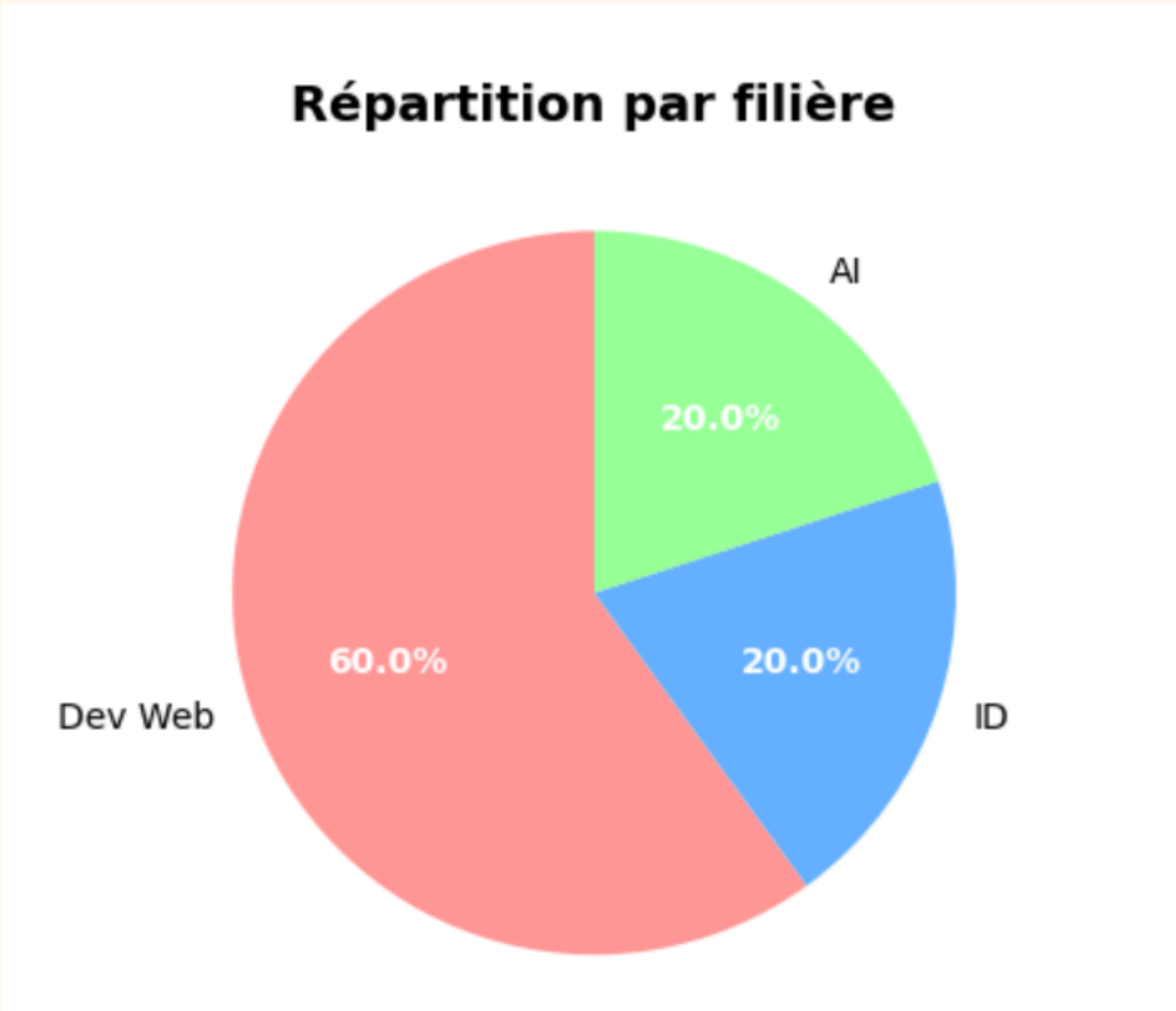
Rechercher:

Nom	Email	Âge	Ville	Filière
Ali	ali@test.com	23	Tanger	Dev Web
Ahmed	ahmed@test.com	25	Rabat	Dev Web
Salma	salma@test.com	27	Casablanca	ID
Sophia	sophia@test.com	19	Tanger	Dev Web
Adam	adam@test.com	19	Rabat	AI

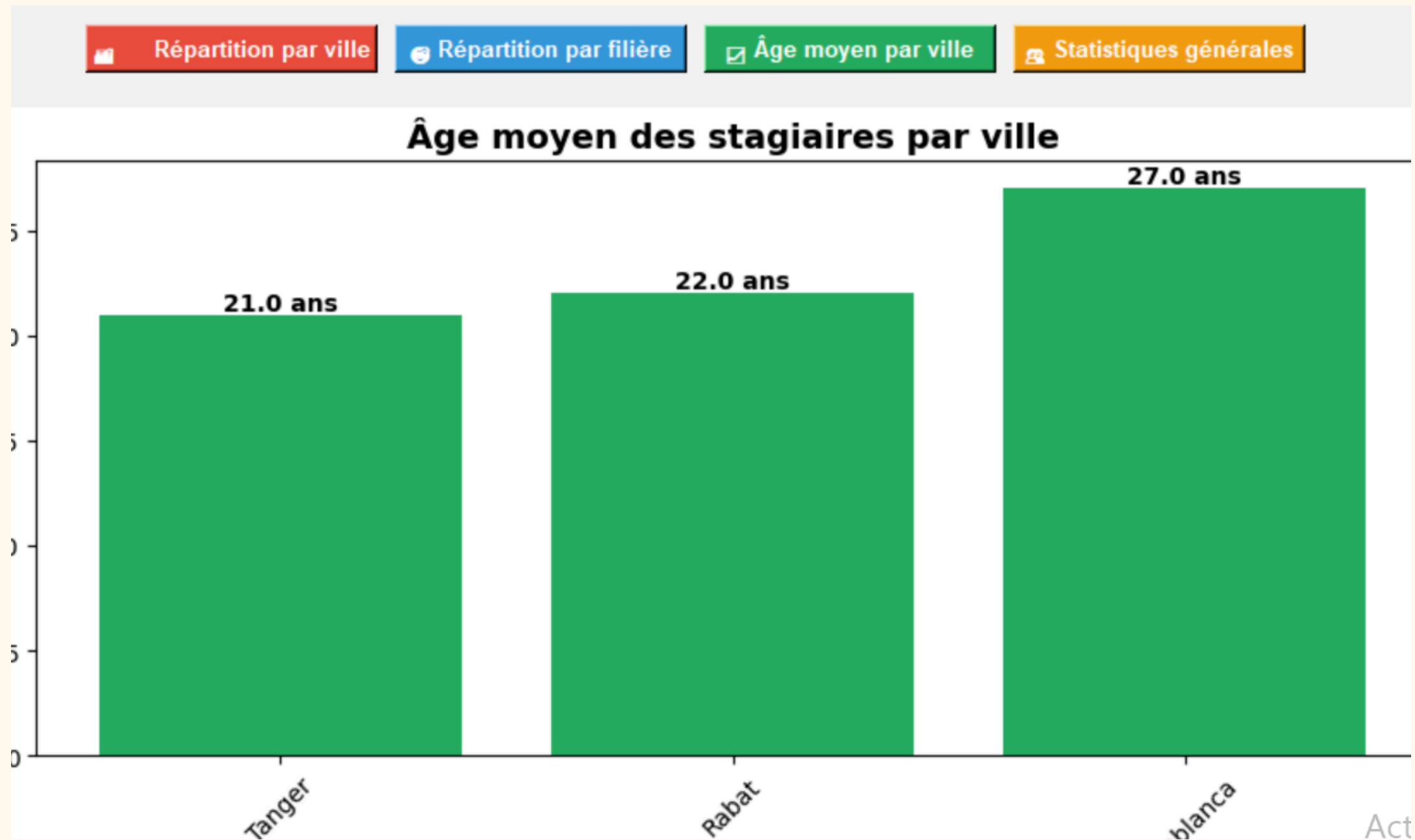
Répartition par ville :



Répartition par filière :



Age moyen par ville :



Statistiques générales :

STATISTIQUES GÉNÉRALES

▢ Total stagiaires: 5

▢▢ Ville la plus représentée:
Rabat
(2 stagiaires)

▢ Filière la plus populaire:
Dev Web
(3 stagiaires)

▢ Âge moyen: 22.6 ans

Agrégation MongoDB par ville :

→ Regroupe les stagiaires par ville, puis compte le nombre.

pipeline = [
{"\$group": {"_id": "\$ville", "total": {"\$sum": 1}}}, {"\$sort": {"total": -1}}]

The screenshot shows the MongoDB Compass interface. On the left, the query editor displays a pipeline aggregation query:

```
1 [
2   {"$group": {"_id": "$ville", "total": {"$sum": 1}}}
3   {"$sort": {"total": -1}}
4 ]
```

On the right, the 'PIPELINE OUTPUT' tab shows a sample of 3 documents. The output is sorted by 'total' in descending order:

- `_id: "Rabat"`
`total : 2`
- `total : 2`
`_id: "Tanger"`
- `_id: "Casablanca"`
`total : 1`

The 'OUTPUT OPTIONS' dropdown is visible in the top right corner of the output panel.

Agrégation MongoDB par filière

→ Produit les données pour un camembert

```
[{"$group": {"_id": "$filiere", "total": {"$sum": 1}}}]
```

The screenshot shows the MongoDB Compass interface. On the left, the aggregation pipeline is defined in a code editor with line numbers 1 to 4. The pipeline consists of a single \$group stage. On the right, the 'PIPELINE OUTPUT' section displays a sample of 3 documents. Each document has a 'total' field (value 1) and an '_id' field. The first two documents have '_id' values of 'ID' and 'AI' respectively. The third document has an '_id' value of 'Dev Web' and a 'total' value of 3.

```
1 [
2   {"$group": {"_id": "$filiere",
3               "total": {"$sum": 1}}}
4 ]
```

PIPELINE OUTPUT
Sample of 3 documents

total : 1
_id: "ID"

total : 1
_id: "AI"

_id: "Dev Web"
total : 3

Âge moyen par ville :

```
{"$group": {"_id": "$ville", "age_moyen": {"$avg": "$age"}}
```

```
1  [
2  {"$group": {"_id": "$ville",
3             "age_moyen": {"$avg": "$age"}}}
4  ]
```

PIPELINE OUTPUT

Sample of 3 documents

OUTPUT OPTIONS ▼

_id: "Rabat"
age_moyen : 22

_id: "Tanger"
age_moyen : 21

_id: "Casablanca"
age_moyen : 27